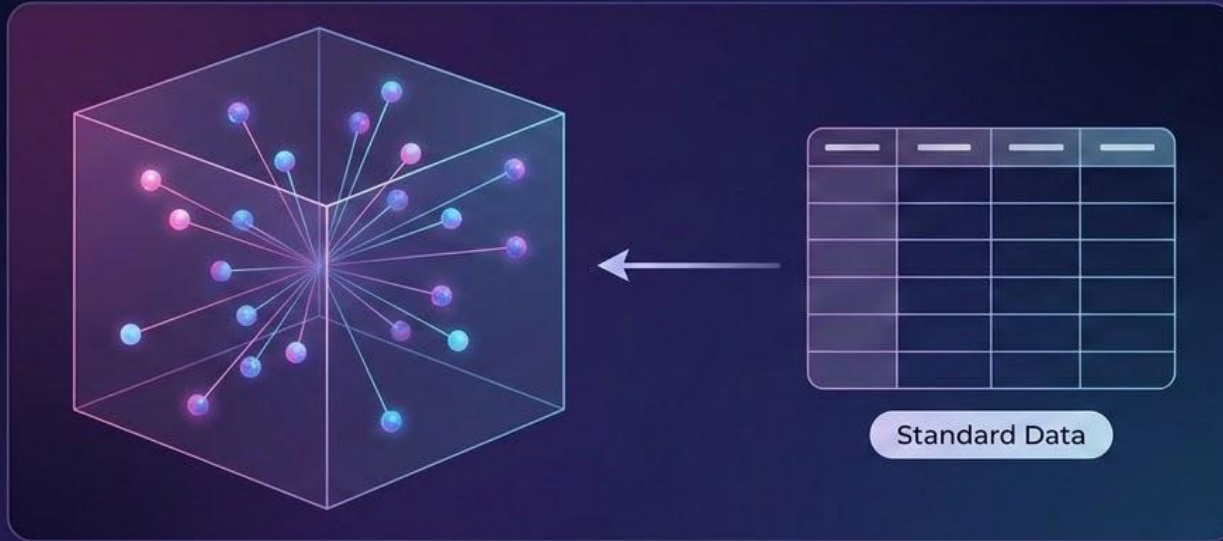


Understanding Vector Databases & Building Simple RAG

Using OpenAI Embeddings, LLM &
Prompt Engineering

What is a Vector Database?

High-dimensional embeddings for semantic search



Enables similarity-based search across vast datasets

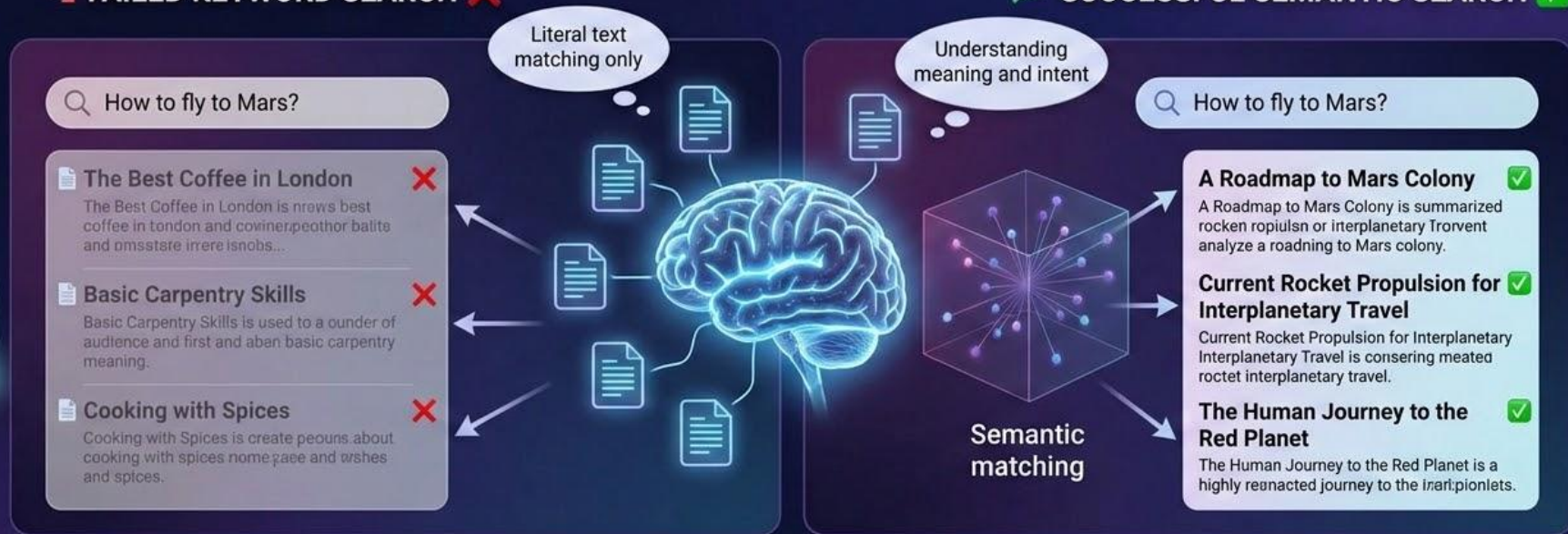


Processes complex, unstructured data (text, images, video)

Why Do We Need Vector Databases?

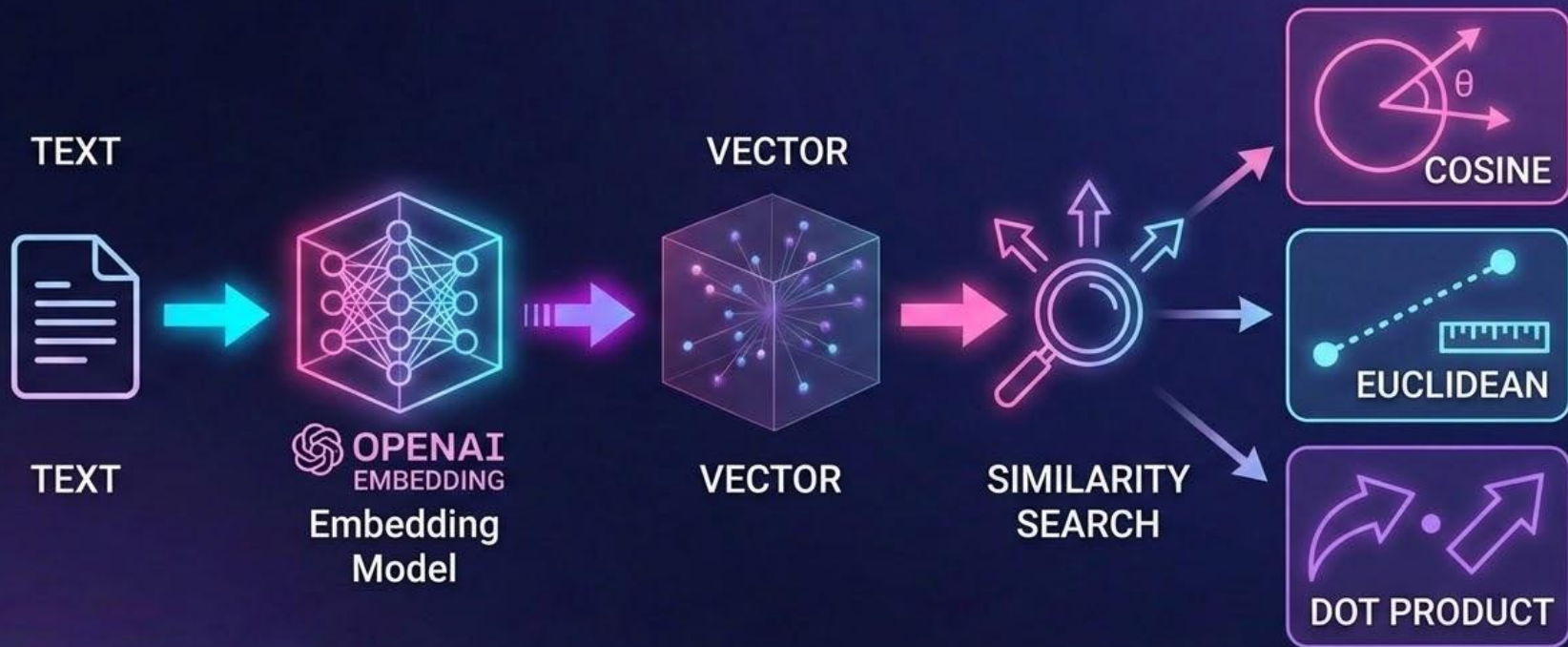
❌ FAILED KEYWORD SEARCH

✅ SUCCESSFUL SEMANTIC SEARCH



- Overcomes semantic gap
- Gives LLMs long-term memory

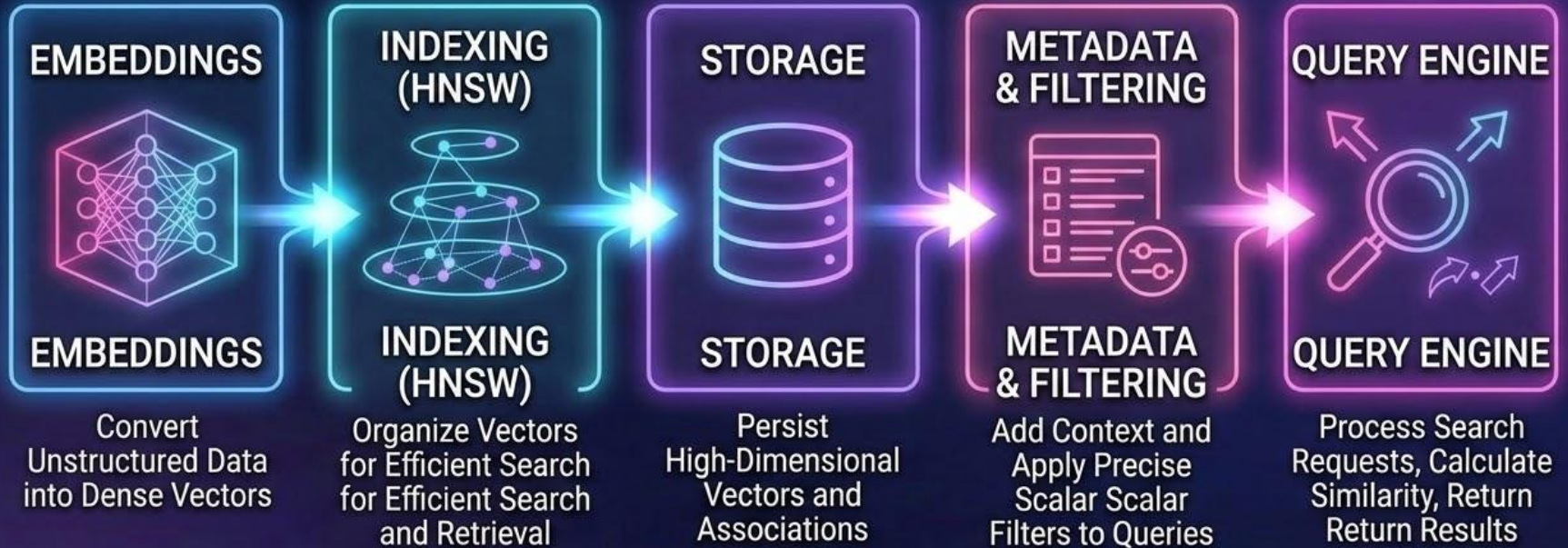
How Vector Databases Work



Vector Database vs Traditional Database

Comparison Point	Traditional Database (SQL Rows)	Vector Database (Glowing Vectors)
Data Type	 Structured Data Tables, schema, text, numbers	 Unstructured Data Images, text embeddings, audio, video
Search Method	 Exact Keyword Matching Finds literal matches; 'Mars' only matches 'Mars'	 Semantic Search & Similarity Finds conceptual matches; 'How to fly to Mars' matches related concepts
Query Processing	 Process Rows Sequentially Traditional processing flow	 Highly Parallel Calculations Uses vector arithmetic and parallel processing for speed
Storage Format	 Flat Tables & Fields Schema-defined data storage	 High-Dimensional Vectors Embeddings as coordinates in vector space
Speed & Scalability	 Fast for Simple Reads Scales vertically by adding resources to one machine	 Blazing Fast for Semantic Tasks Scales horizontally across clusters of machines for high performance

Key Components of a Vector Database



Popular Vector Databases (2026)



Chroma

Strength: AI-native open-source embedding database for rapid development.



Pinecone

Strength: Fully managed, serverless vector database for easy, scalable applications.



Weaviate

Strength: Flexible, cloud-native open-source database for structured and unstructured data.



Qdrant

Strength: Lightning-fast, highly-performant vector database built in Rust for efficiency.



Milvus

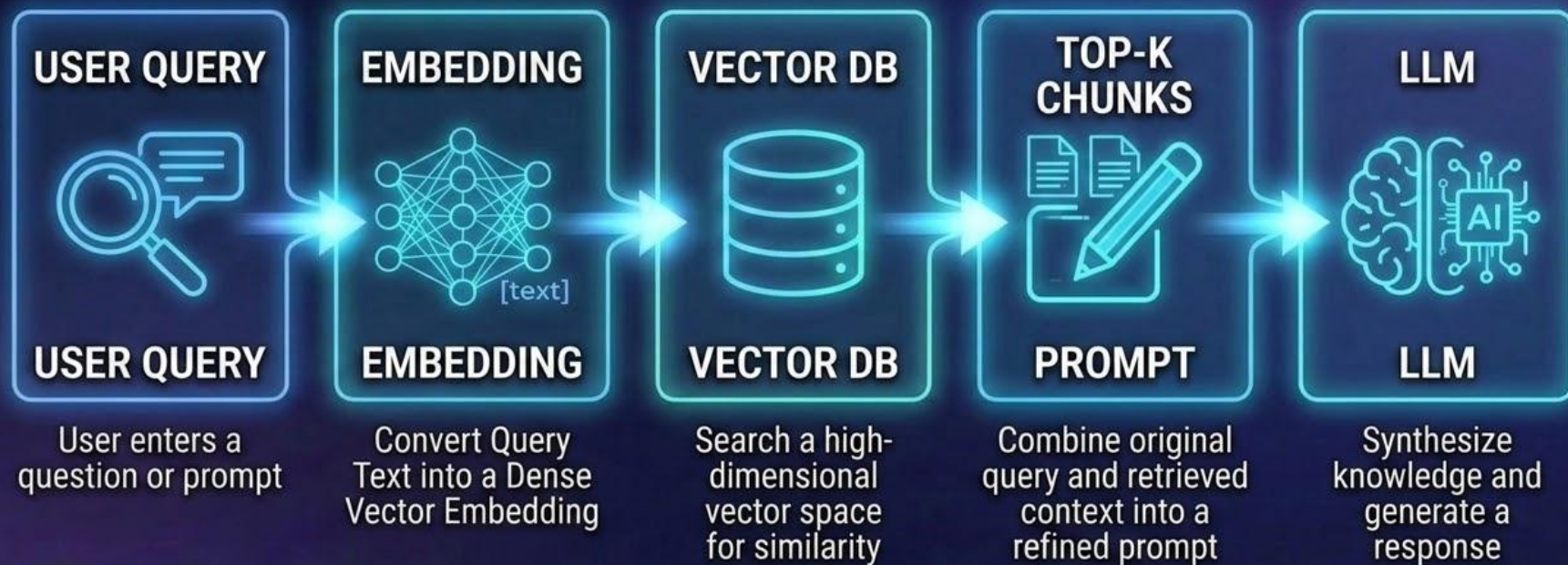
Strength: Mature, cloud-native open-source vector database built for massive scale and reliability.



PGVector

Strength: Open-source extension adding vector search capability seamlessly to PostgreSQL.

Retrieval Augmented Generation (RAG)



Real-World Use Cases & Future



Chat with Docs

Analyze and query large internal document repositories.



Semantic Search

Understand intent beyond keywords to deliver precise information.



AI Agents

Build proactive AI assistants that execute complex multi-step tasks.



Recommendations

Personalize user experiences with hyper-relevant suggestions.



Legal AI

Automate contract review, discovery, and case law analysis.



Education

Create tailored learning paths and personalized tutoring.

THANK YOU!

QUESTIONS?

Let's Build Simple RAG with OpenAI + Chroma



Let's discuss...

[Insert Your Name Here]